

IMAD EDDINE EL MOUSS

Research Engineer | On-Premises Industrial AI | ML Systems & Data Platforms

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PROFILE

Research engineer building industrial AI systems that run inside the plant rather than in the cloud. I own the event-driven platform that carries a Finnish research group's manufacturing pilots, ship local LLM and computer-vision pipelines against real partner data, and hold the evaluation discipline that decides whether a prototype is genuinely deployable. Named inventor on a European patent application and a contributor to a EUR 7M Horizon Europe consortium.

SELECTED HIGHLIGHTS

- **Patent pending.** Named inventor (one of three) on a European patent application filed at the EPO on 10 April 2026: adaptive digital content generation based on multimodal data. Applicant: Turku University of Applied Sciences.
- **EUR 6.98M Horizon Europe project.** Contributed to the planning of *Sensor4Food* (HORIZON-CL4), a 21-partner European consortium coordinated by Turku UAS, developing photonic sensing and AI for the poultry value chain across pilot sites in Spain, Italy and Norway (2026–2030).
- **Platform ownership.** Architect and principal developer of AIOP, the on-premises event-driven AI platform behind every pilot module in the research group. Reproducible stress audits: 200 events/s sustained at 0% loss, 10 ms p99 latency.
- **Technical lead for the group's student engineering portfolio** — more than twenty applied-AI R&D projects across a rotating cohort, spanning 3D reconstruction, annotation tooling, speech-to-text, data anonymisation and ERP extraction.

EXPERIENCE

Research Engineer

Cognitive Technologies Research Group (CoRe), Turku University of Applied Sciences

Jan 2026 – Present

Turku, Finland

- Designed and built **AIOP**, an on-premises event-driven AI platform in Python (NATS JetStream, Docker, FastAPI): immutable event envelope, plugin SDK, single-writer state projector, mandatory per-site data isolation, health and metrics endpoints. It is the integration backbone for every industrial pilot the group runs.
- Established the platform's performance evidence through a reproducible pair of GPU-workstation stress audits — 75 and 200 events/s sustained at **0% drop**, p99 of 5 ms and 10 ms, measured ceiling ≈ 260 events/s. Removed an $O(N)$ state-staging bottleneck via a touched-entity transaction overlay, lifting a projected ≈ 40 events/s ceiling at million-entity scale, and shipped a durable SQLite plugin outbox so events survive broker outages.
- Deliver partner pilots across two funded programmes — **TeoÄly** (ERDF, $\approx$ EUR 320k) and **ADINO** (Business Finland, ten industrial partners): machine-vision surface and paint inspection, spectral sensing for food-process control, document and ERP extraction, speech-to-report.
- Own the sensing track of a food-process pilot after it moved off video: specified 14-channel AS7343 spectral sensors over I²C on Raspberry Pi and ESP32-class hardware, defined the versioned event contract they publish, and merged the classification plugin that turns colour samples into operator-facing process recommendations.
- Local-first LLM and VLM engineering: quantised inference with llama.cpp and vLLM on RTX 5090/3090 workstations, schema-enforced structured extraction, and speech-to-report pipelines (whisper-large-v3-turbo int8 into a quantised instruct model) where a hallucinated field is treated as an extraction failure, not an acceptable output.
- Shipped a privacy-bounded desktop AI assistant for partner offices: local inference endpoint only, a startup egress guard that blocks any non-configured destination, cloud session sync disabled, released as Linux and macOS installers through self-hosted CI at zero hosted-runner cost.
- Provide technical supervision across the group's intern cohort — more than twenty R&D projects to date on a rolling intake, with several running concurrently at any time. Scoping, method choice, GPU workstation provisioning, code review, and the handoff standard now used to judge every project: runnable setup, declared output schema, evaluation evidence and documented limitations.
- Authored the group's **EU AI Act** position for its industrial offerings: per-offering Annex III exposure, the timeline as amended by Regulation (EU) 2026/1744, and task-level rather than person-level monitoring as a default design rule.

Applied AI Intern — Horizon Europe proposal work

Cognitive Technologies Research Group (CoRe), Turku University of Applied Sciences

Apr 2025 – Aug 2025

Turku, Finland

- Built a multi-source extraction prototype combining OCR, document parsers and language models to turn heterogeneous industrial documents into schema-validated structured output.
- Contributed privacy-preserving data architecture, ontology alignment and multimodal harmonisation to a Horizon Europe proposal, and shaped the data-hub concept linking extraction, standardisation and downstream AI consumption.

Research Intern — Autonomous & Intelligent Systems

Feb 2025 – Jun 2025

Turku University of Applied Sciences

Turku, Finland

- Built ETL enrichment workflows for time-series sensor data in InfluxDB and applied anomaly detection, forecasting and classification to production telemetry.
- Automated backup and recovery for production services across the TIG stack and PostgreSQL, working on both operational reliability and the analytics that ran on top of it.

SELECTED PROJECTS

IPKE — Procedural Knowledge Graph Extraction from Industrial Documentation

B.Eng. thesis, graded 5/5

Privacy-by-design local LLM pipeline that turns safety-critical industrial manuals into auditable knowledge graphs of steps, constraints and decision points. Hierarchy-aware semantic chunking, two-stage prompting, a purpose-built constraint-attachment metric, and a multi-seed evaluation harness with bootstrap confidence intervals and inter-annotator agreement gating.

NaviCast — Real-Time Maritime Intelligence Pipeline

Python, MQTT, PostgreSQL, FastAPI

Live AIS ingestion and analytics system with forecasting models and an API layer for operational decision support on vessel movement.

Federated Learning Privacy Benchmark

PyTorch, reproducible evaluation

Reproducible benchmark quantifying gradient-inversion risk and the measurable defensive value of privacy-preserving training methods.

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript, Bash
ML & Vision	PyTorch, scikit-learn, TensorFlow, YOLO-family detectors, VLM prompting and LoRA/QLoRA fine-tuning, anomaly detection, forecasting
LLM Systems	llama.cpp, vLLM, Ollama, quantisation, schema-enforced extraction, RAG, knowledge graphs, Whisper STT, evaluation-harness design
Data & Pipelines	NATS JetStream, MQTT, Kafka, Airflow, ETL, PostgreSQL, InfluxDB, Neo4j, validation and row-level provenance
Platform & Ops	Docker, FastAPI, Linux, Git, self-hosted CI, Grafana, NVIDIA RTX 5090/3090 workstations, Raspberry Pi and ESP32 edge hardware
Practice	On-prem and privacy-by-design architecture, reproducible benchmarking, EU AI Act risk classification, technical supervision, documentation and handoff standards

EDUCATION

B.Eng. Data Engineering & Artificial Intelligence (ICT)

Jan 2023 – Dec 2025

Turku University of Applied Sciences

Turku, Finland

Graduated with 243 ECTS against a 240 ECTS programme. Degree average 4.32/5. Thesis graded 5/5.

RESEARCH OUTPUT, CERTIFICATIONS & LANGUAGES

Output	Named inventor on a European patent application (EPO, April 2026) • contributor to the funded Sensor4Food Horizon Europe proposal • lead author of an applied paper in preparation on running AI systems on-premises in manufacturing
Certifications	AWS Academy — Cloud Foundations, Machine Learning Foundations, Machine Learning for NLP • DataCamp Certified Data Engineer • MATLAB Onramp & Fundamentals
Languages	Arabic (native) • Amazigh (native) • English (full professional) • French (native) • Finnish (A2)